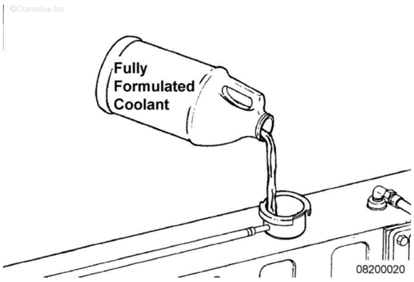
Trainings

General Information

Water

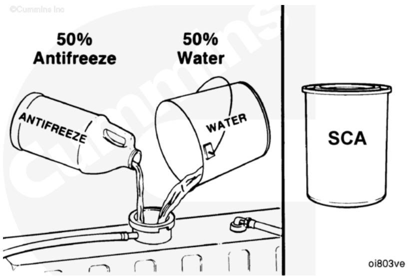
Always use good-quality soft water in the coolant mixture. Water added to the cooling system **must** meet the specifications given in the accompanying chart.

Mineral	Problem Caused	Maximum Limits
Calclum Magnesium (hardness)	Deposits on Liners/Heads/Coolers	170 ppm
Chloride	General Corrosion	40 ppm
Sulfate	General Corrosion	100 ppm



⚠ CAUTION ⚠

Antifreeze over concentration reduces protection. Do not use more than 68-percent antifreeze or overheating can result. A mixture of 50-percent antifreeze and 50-percent water is sufficient for freeze protection to -37°C [-35°F]

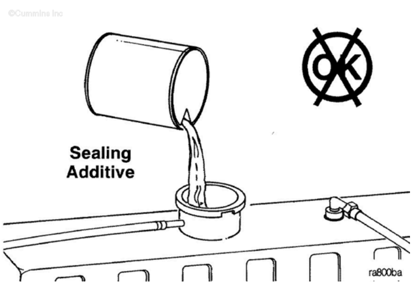


⚠ CAUTION ⚠

Do not use a high-silicate antifreeze. A silicate-gel (hydro-gel) formation can occur when a cooling system contains an over concentration of high-silicate antifreeze and/or supplemental coolant additives. Engine damage will result.

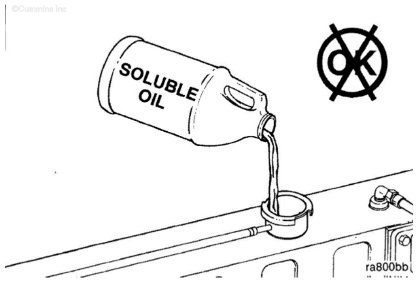
Use ethylene-glycol or propylene-glycol antifreeze year-round to provide freezing and boil-over protection.

- Buildup in coolant low-flow areas
- Clogged coolant filters
- Plugged radiator.



Do **not** use soluble oils in the cooling system. The use of soluble oil will:

- Allow cylinder liner pitting
- Corrode brass and copper
- Damage heat-transfer surfaces
- Damage seals and hoses.



Supplemental Coolant Additive (SCA)

Supplemental coolant additives, SCA, or equivalent, are required to protect the cooling system from fouling, solder blooming, liner pitting, and general corrosion. The coolant filter is required to protect the cooling system from abrasive material, debris, and precipitated coolant additives.